

Iberian Cantilevers

Overhead line equipment from the Iberian Peninsula
for *Create: Pantographs & Wires*

Feature overview

Minecraft 1.20.1 (Forge) · version 0.1.0 · October 2026

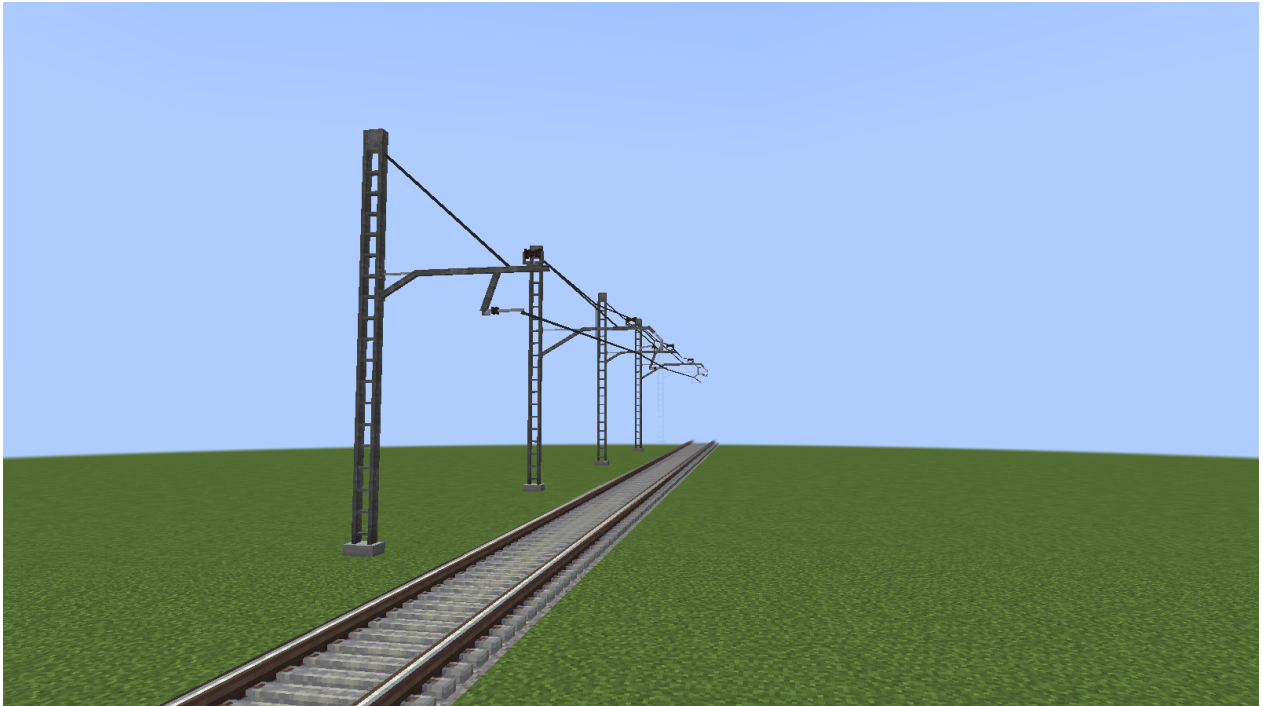


Figure 1: A stretch of line built with the add-on: Iberian flat lattice masts, wide Iberian cantilevers in zigzag held by diagonal stays, and the overhead wire of *Create: Pantographs & Wires*.

Developed by a railway enthusiast.

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1 Overview

Iberian Cantilevers is an add-on for *Create: Pantographs & Wires* (PNW) that adds overhead line equipment as it is built on the railways of the Iberian Peninsula, mostly following Spanish practice: the cantilevers (*ménsulas*) of the conventional network, the masts that carry them, the rigid overhead conductor rail used in tunnels and an Iberian-style pantograph.

The add-on does not replace any part of PNW. It plugs into PNW's wire system: its blocks are PNW wire connectors, the catenary wire of PNW is strung onto them as usual, pantographs collect current from them in the same way, and the masts share PNW's tags, so PNW blocks and Iberian blocks can be mixed freely on the same line.

Contents of the add-on

- **Iberian Cantilever** – a configurable cantilever modelled on the Spanish one, with three zigzag positions, three insulator types, widths from 1.5 to 6.5 blocks, an optional support wire and, on wide cantilevers, a diagonal stay to the mast (Section 3).
- **Iberian masts** – flat lattice (straight and diagonal) and square lattice masts with four weathering stages and waxing (Section 4).
- **Rigid catenary** – an overhead conductor rail strung like a wire, which follows curves and can be joined to the flexible catenary (Section 5).
- **Iberian Tunnel Support** – ceiling and wall supports for the rigid catenary in several variants, with an overall size setting that enlarges the support and its profile (Section 6).

- **Iberian Pantograph** – PNW’s pantograph with the red frame typical of Iberian trains (Section 7).

2 Real-world reference

On the Spanish conventional network the contact wire is usually held by *cantilevers*: a steel tube fixed to a lattice or H-beam mast, which rises in a short diagonal and then runs horizontally over the track. The messenger wire rests on an insulator on top of the tube, and a registration arm hanging from the cantilever pulls the contact wire alternately to each side of the track (*zigzag*), so that the pantograph strip wears evenly. A support wire (stay) often runs from the mast to the elbow of the tube.



(a) Cantilevers on lattice masts, Sant Adrià de Besòs.

Adif – Ineco, CC BY-SA 3.0, via Wikimedia Commons.



(b) Commuter train under cantilevers on lattice masts, Sant Adrià de Besòs.

Smiley.toerist, CC BY-SA 4.0, via Wikimedia Commons.

Figure 2: Cantilevers of the Spanish conventional (Iberian gauge) network.



(a) Conventional line with cantilevers on lattice masts, Nules (Valencia–Castellón).

Falk2, CC BY-SA 4.0, via Wikimedia Commons.



(b) Detail of a cantilever with its insulators and registration arm, València Font de Sant Lluís.

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Figure 3: Conventional catenary in the Valencia area.

In tunnels and underground stations the flexible catenary is often replaced by a *rigid catenary*: an aluminium or steel profile that clamps the contact wire and hangs from short supports fixed to the ceiling or to the tunnel wall. It needs very little height and almost no maintenance of tension.

Many Spanish electric trains carry pantographs with a frame painted red, which is where the colour of the Iberian Pantograph comes from.



Figure 4: Rigid catenary and its ceiling supports, Nuevos Ministerios (Madrid Metro, Line 10).

Draceane, CC BY-SA 4.0, via Wikimedia Commons.



(a) Pantograph on a Renfe Class 446 train, Córdoba.

Philip Mallis, CC BY-SA 4.0, via Wikimedia Commons.



(b) Pantograph on a Renfe Class 449 train.

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Figure 5: Red pantograph frames on Spanish trains.

3 Iberian Cantilever

The Iberian Cantilever is placed on the side of a mast, exactly like PNW's cantilever, and accepts the PNW catenary wire: the messenger wire is attached on top of the insulator and the contact wire at the tip of the registration arm. All its parts are built from separate models (tube, perforated drop pieces, registration arm, insulators, support wire) that are assembled and sized automatically from its settings, so the cantilever always reaches exactly where the wires are.

3.1 Zigzag position

- **Inner** – the contact wire is held towards the mast. The registration arm hangs from the diagonal or, for wide cantilevers, from an L-shaped strut below the horizontal tube.
- **Middle** – the tube continues past the messenger insulator, drops at 45° into a perforated piece and a longer registration arm descends to the wire.
- **Outer** – the tube drops into a diagonal and a vertical perforated piece, and the registration

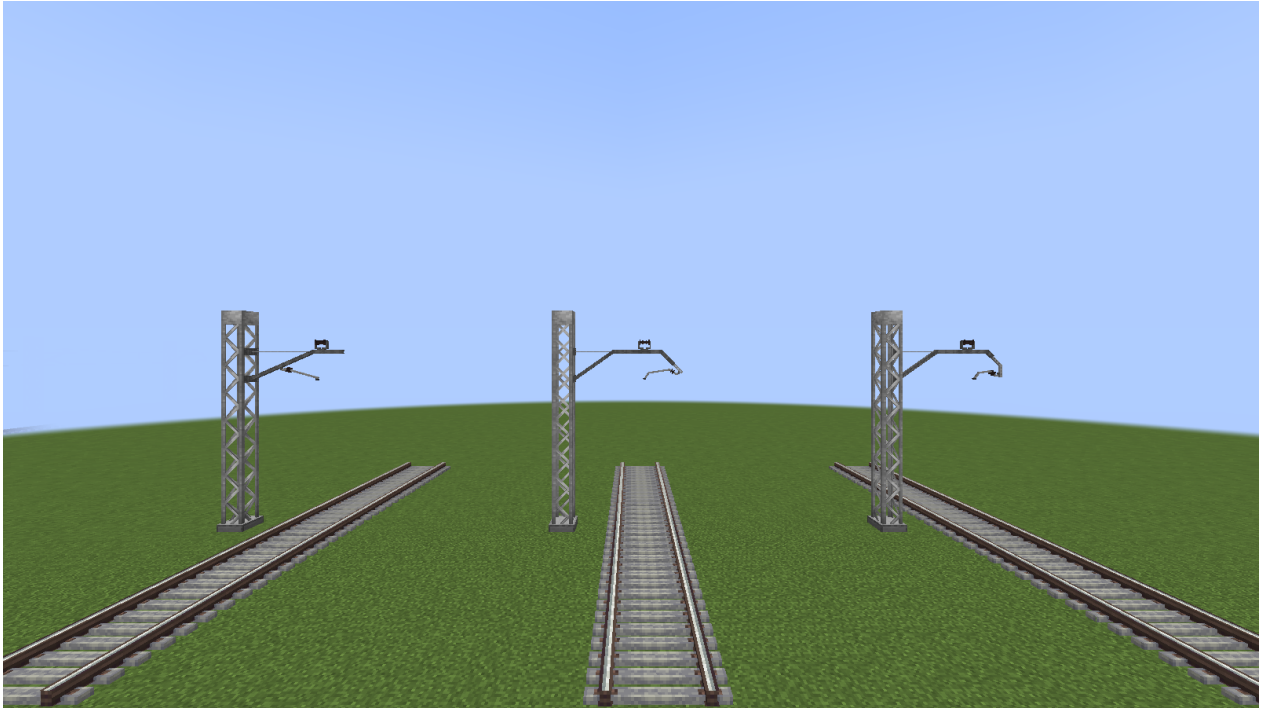
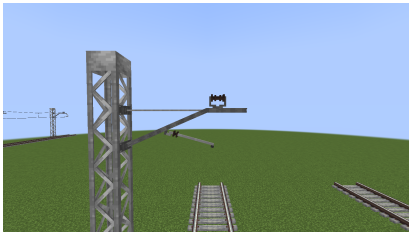
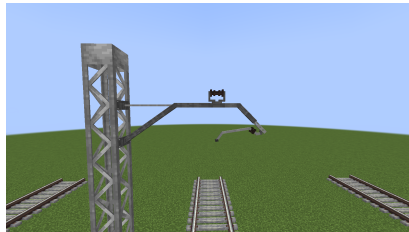


Figure 6: The three zigzag positions of the contact wire: inner, middle and outer.

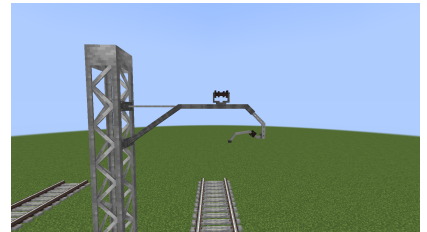
arm points back towards the mast.



(a) Inner.



(b) Middle.



(c) Outer.

Figure 7: Side view of each zigzag position. The middle and outer positions use an arched registration arm with its own insulator.

3.2 Insulator type and support wire

The insulator that carries the messenger wire can be chosen among three types, and the support wire (stay) between the mast and the elbow of the tube can be added or removed. Every insulator type works with every zigzag position: the nine combinations are shown below.

- **Horizontal support** – the messenger rests on top of a horizontal insulator.
- **Horizontal hanging support** – the insulator hangs below the tube and the messenger passes through it (shown without support wire).
- **Vertical support** – a vertical insulator on top of the tube.

3.3 Width

The width is the distance from the mast to the track axis, from 1.5 to 6.5 blocks in one-block steps, as with PNW's cantilever. The tube, the diagonal and the registration arm are sized automatically

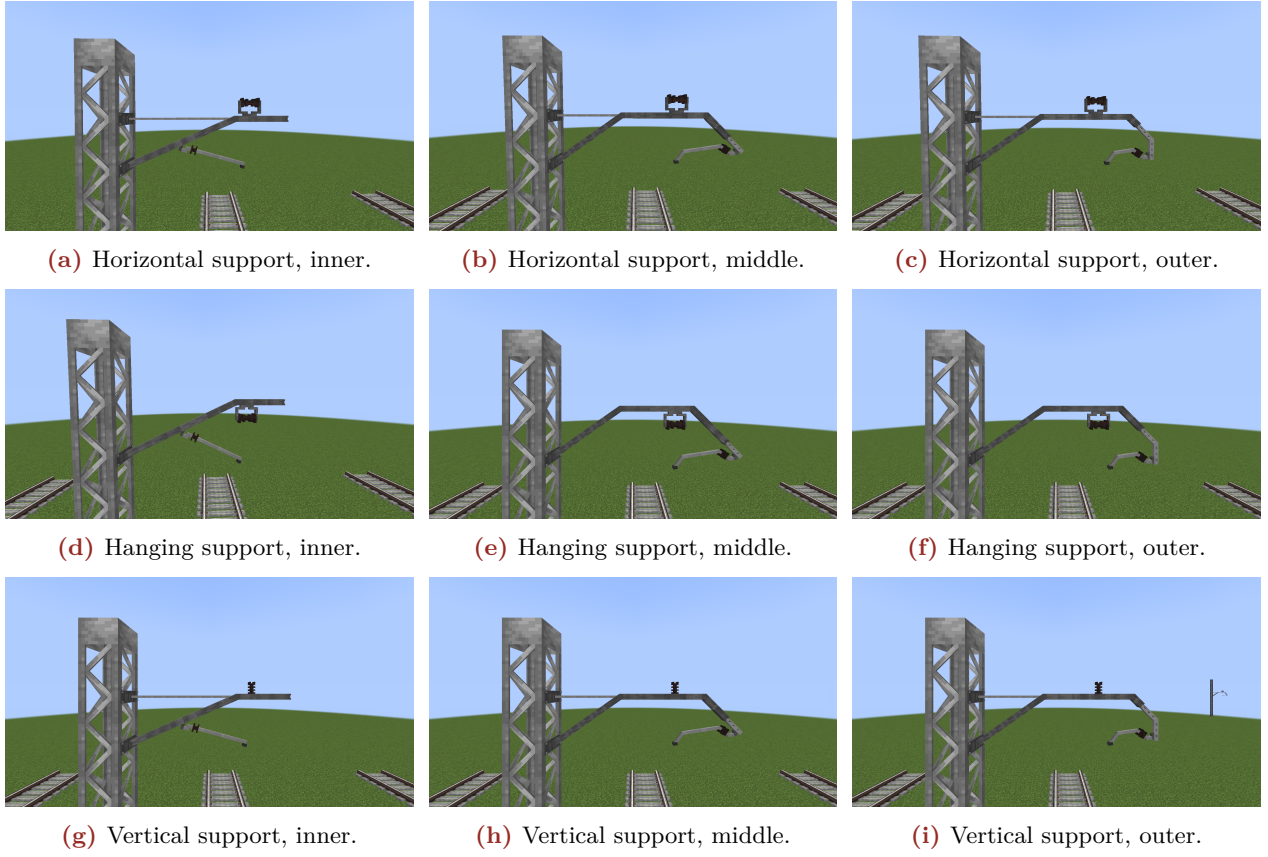


Figure 8: All nine combinations of insulator type (rows) and zigzag position (columns).

for each width, so the messenger always sits above the track and the contact wire at its zigzag position.

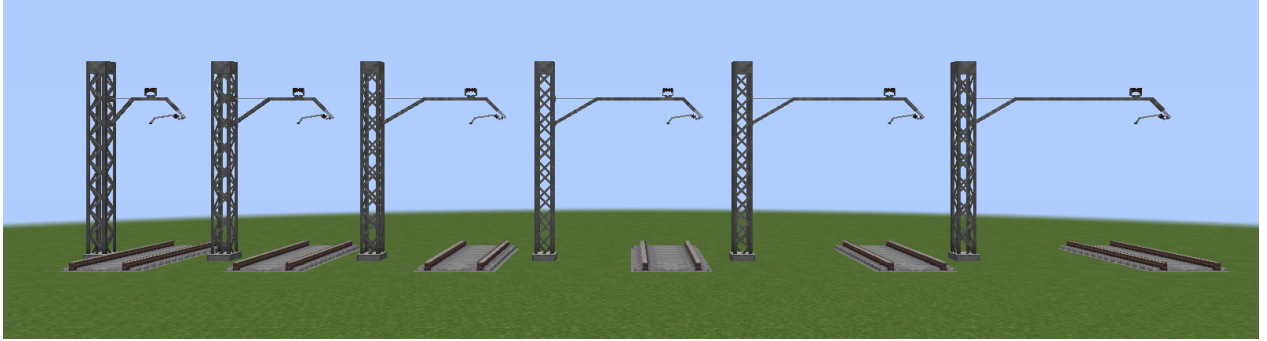


Figure 9: The six widths, from 1.5 (left) to 6.5 blocks (right), on Iberian square lattice masts.

3.4 Diagonal stay

Wide cantilevers (width 3 or more) can be hung from the mast with a *diagonal stay*. With PNW's wire coil set to *Support Wire*, click the cantilever and then a block of the mast between 2 and 5 blocks above it (or the other way round). The stay is a single straight wire that enters the horizontal tube just before the messenger insulator and the face of the mast that looks towards the cantilever, so it looks fastened to the steel at both ends.

- The attachment point on the tube moves outwards with the width, so the stay always holds the far end of the cantilever.
- The height on the mast is chosen by the block that is clicked (2 to 5 blocks above the can-

tilever), which gives different triangles.

- If the cantilever is reconfigured the stay follows it; if it becomes narrower than 3 blocks the stay is removed. Breaking the cantilever or that mast block returns the wire to the coil.

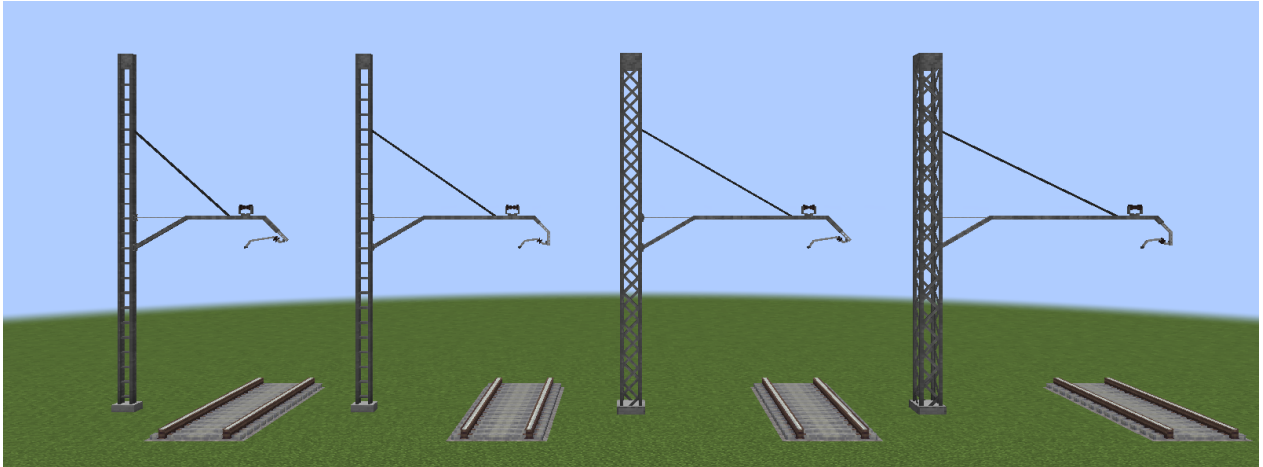


Figure 10: Diagonal stays on cantilevers of width 3.5, 4.5, 5.5 and 6.5, all to a mast block 3 blocks higher: the attachment on the tube moves outwards with the width.

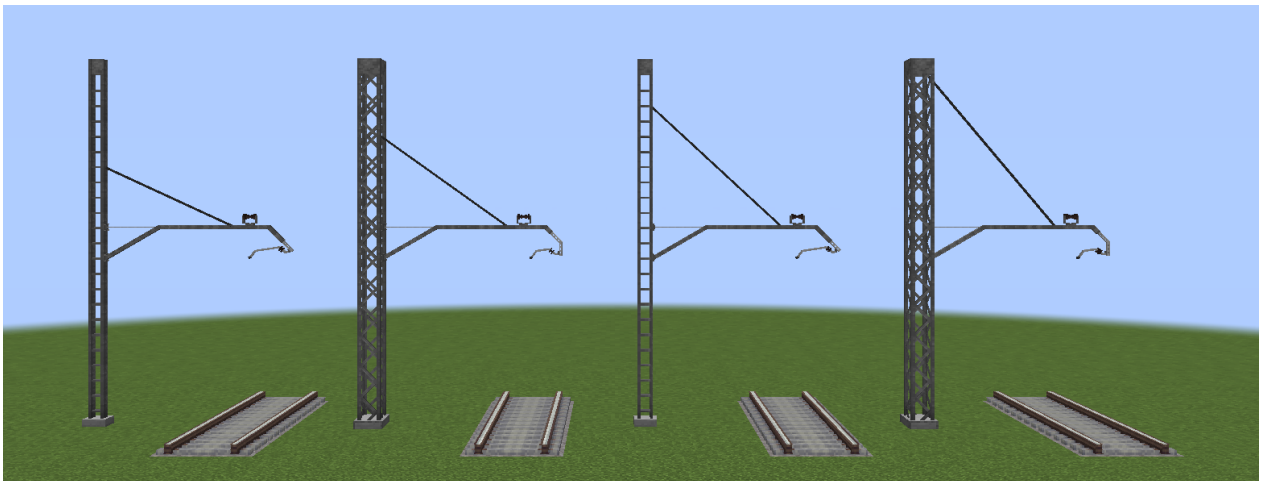
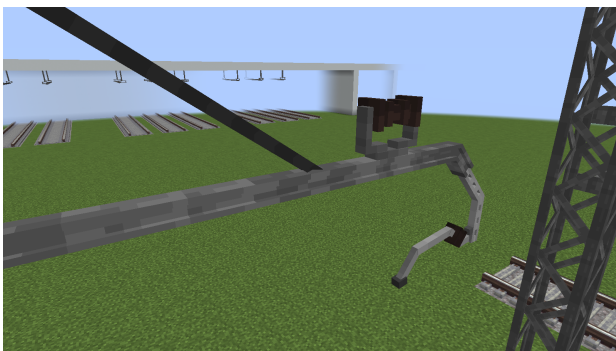
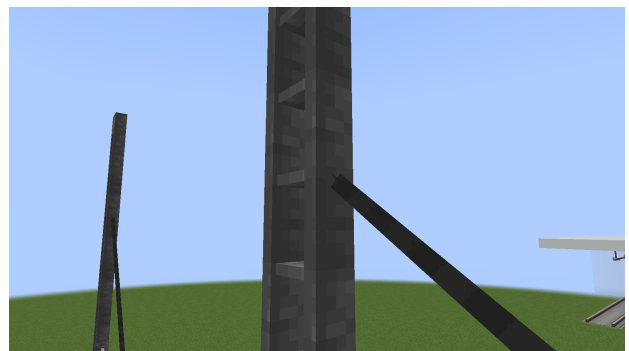


Figure 11: The same cantilever (width 4.5) with the stay 2, 3, 4 and 5 blocks up the mast.



(a) The stay enters the horizontal tube before the messenger insulator.



(b) The stay enters the face of the mast.

Figure 12: Both ends of the diagonal stay.

3.5 Settings window

Right-clicking with the cantilever item in hand (or with an empty hand on a placed cantilever) opens a settings window in the same style as PNW's, with a live 3D preview.

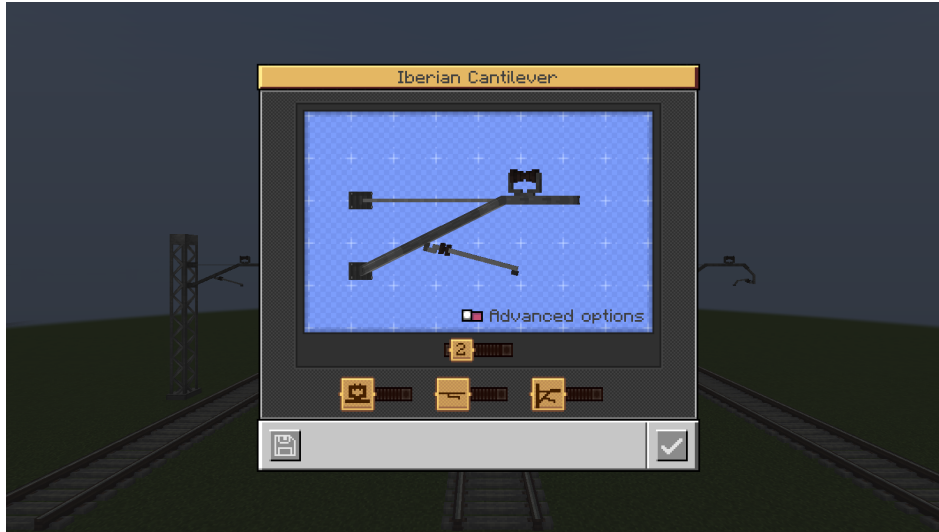


Figure 13: Settings window of the Iberian Cantilever.

Setting	Effect
Width	Reach of the cantilever from the mast to the track axis, from 1.5 to 6.5 blocks.
Insulator type	Horizontal support, horizontal hanging support or vertical support.
Contact wire	Zigzag position: inner, middle or outer.
Support wire	With or without the stay between mast and tube.
Supporter height	Drop of the diagonal of the tube (advanced).
Catenary height	Height of the contact wire below the block (advanced).
Y offset	Lowest the whole cantilever (advanced).
Set default config	Stores the current settings as the default for new, unconfigured items.

Without advanced options the heights follow fixed proportions, as PNW does with its own cantilever. Changing the settings of a cantilever that already has wires attached moves the wires (and the diagonal stay) with it.

3.6 Placement

- Attaches to PNW masts, Iberian masts, concrete pillars and any other block that PNW's cantilever accepts, in the same 16 directions as PNW blocks (including 45° on diagonal masts).
- The mounting plate adapts to the thickness of the mast and widens on thick masts or full blocks without deforming its bolts.
- The item stacks up to 64.

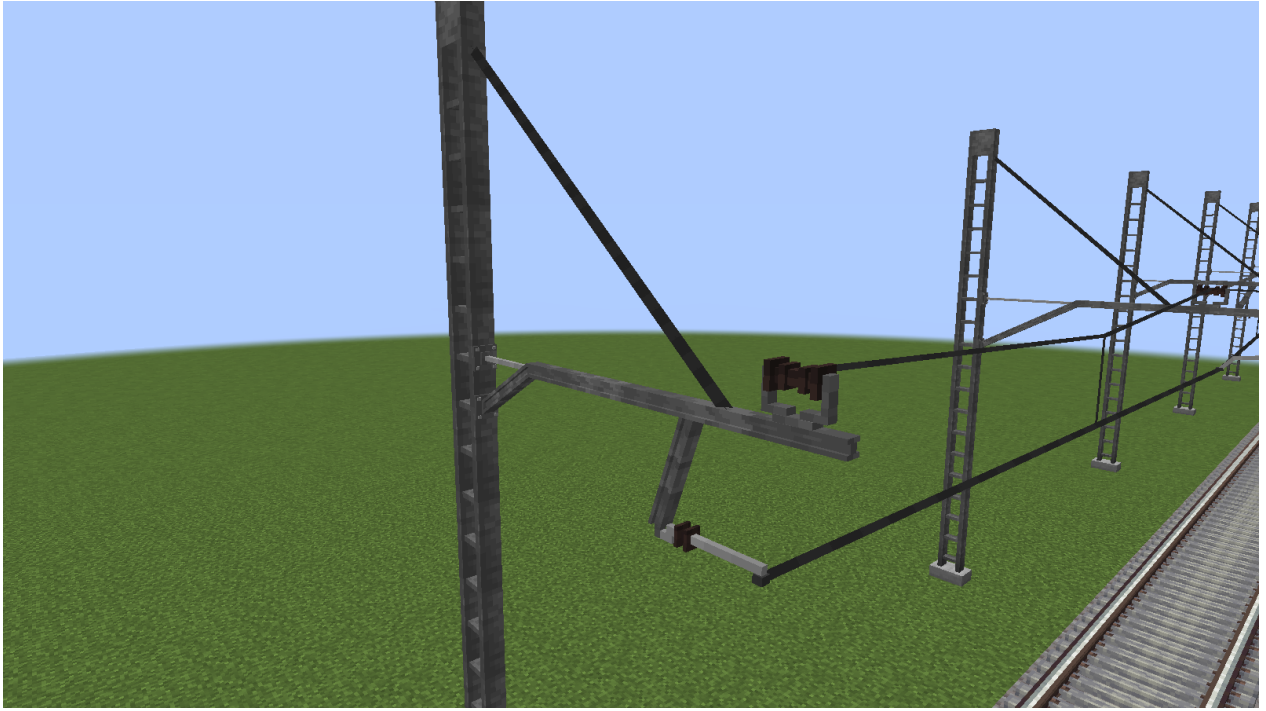


Figure 14: An outer cantilever strung with PNW catenary wire (messenger, contact wire and droppers), with its support wire and diagonal stay.

4 Iberian masts

Three mast families with an Iberian finish: *Iberian Flat Lattice Mast*, *Iberian Flat Diagonal Lattice Mast* and *Iberian Lattice Mast* (square section). They reuse PNW's mast behaviour:

- four weathering stages (new, exposed, weathered, oxidized) that progress over time;
- waxing with honeycomb and wax removal with an axe, as with PNW masts;
- the same connection tags as PNW masts, so both PNW's cantilevers and Iberian Cantilevers attach to them, and the flat mast can be switched between its straight and diagonal versions by crafting.

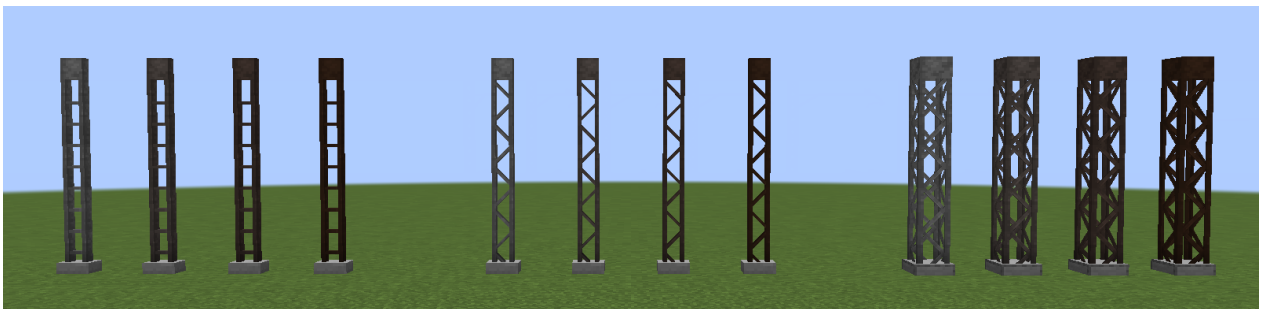


Figure 15: The three Iberian masts – flat lattice, flat diagonal lattice and square lattice – each in its four weathering stages: new, exposed, weathered and oxidized.

5 Rigid catenary

The *Rigid Catenary Profile Bundle* is used like PNW's wire coil: click a support, then the next one, and a rigid profile is laid between them. For PNW it is one more wire of the catenary network, so pantographs collect current from it.

- A bundle holds 200 metres of profile and is consumed by length; a bar shows what is left.
- Where two spans meet at a support with a gentle angle, the profile bends smoothly through the support instead of making a sharp corner.
- Breaking a span returns its metres to the bundles in the player's inventory; whatever does not fit is dropped as a single bundle.

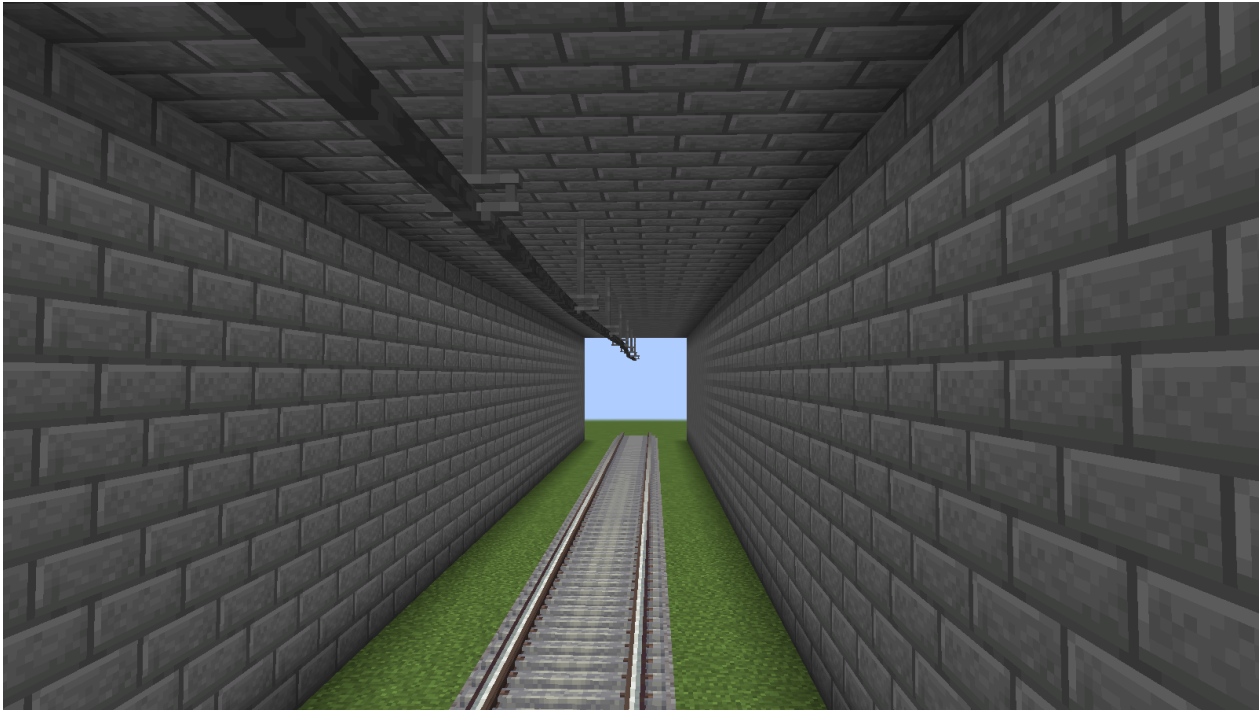


Figure 16: Rigid catenary through a tunnel, with ceiling supports alternating their clamp position to make the zigzag.

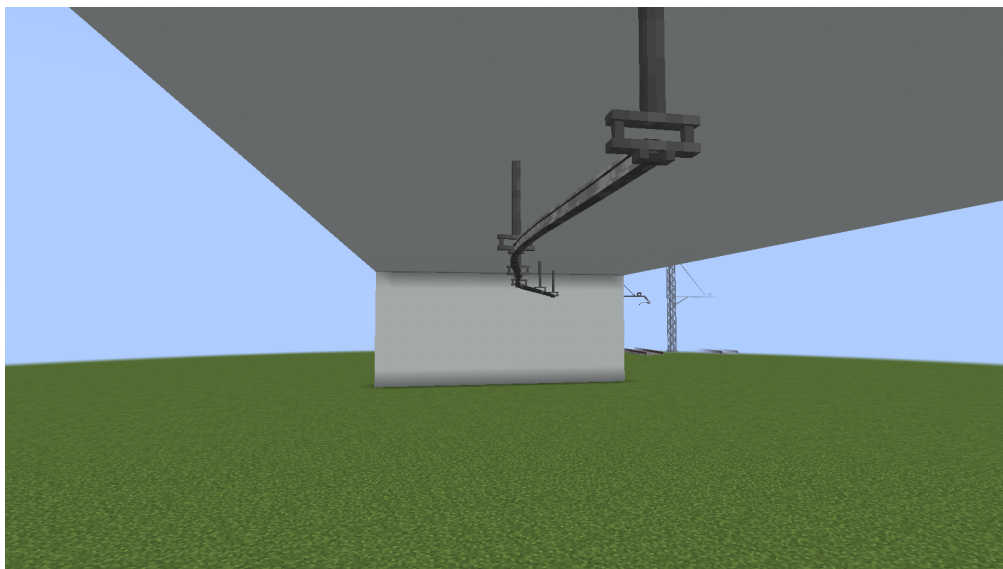
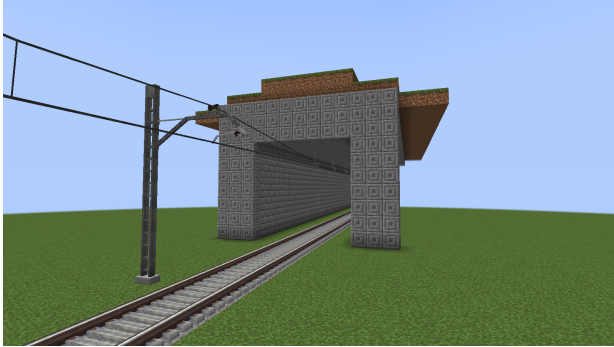


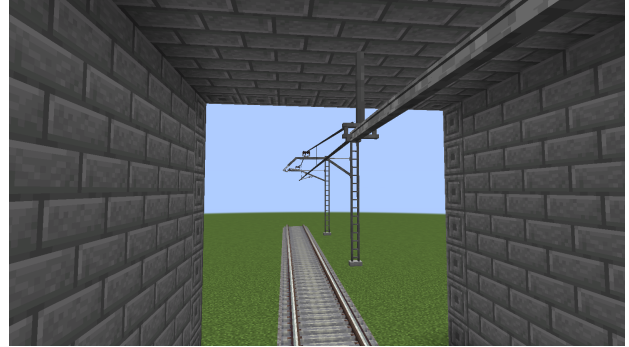
Figure 17: The profile following a curve: between supports it is laid in short straight pieces that bend smoothly through each clamp.

5.1 Transition from flexible to rigid catenary

PNW's catenary wire can be strung from a cantilever to a tunnel support. The contact wire enters the clamp of the rigid profile and the messenger wire descends to an anchoring point just above it; at the transition support the end of the profile turns towards the incoming contact wire so that the pantograph passes smoothly. With default settings, a tunnel support placed under a ceiling at the height of a cantilever keeps the contact wire level across the transition.



(a) The flexible catenary entering the tunnel mouth.



(b) Seen from inside: the rigid profile ends at the first support and the contact wire and messenger continue to the cantilever outside.

Figure 18: Transition between flexible and rigid catenary.

6 Iberian Tunnel Support

A single block covers every support of the rigid catenary. It is configured from one settings window and comes in two versions:

- **Ceiling** – hangs from the block above on a rod. Two sizes (normal and large, with cross-members spanning the whole block) and three clamp positions (left, centre, right) to build the zigzag. Its height can be increased in quarter-block steps; the rod stretches accordingly.
- **Wall** – a horizontal insulated arm fixed to a wall, to the side of a PNW mast or to a concrete pillar, placed like a cantilever in 16 directions. Two arm lengths (short and long). Its height can be moved up or down within the block in half-pixel steps, starting from the centre.

The version is chosen automatically when placing: clicking the side of a wall or pillar places the wall version, clicking the underside of a ceiling places the ceiling version.

6.1 Overall size

The *Overall size* setting (100 %, 125 % or 150 %) makes the whole set larger: the support and the rigid profile strung from it. The ceiling version grows around its clamp, so the contact wire stays at the same height and the rod still reaches the ceiling; the wall version grows from the wall, so its arm becomes longer. Each span of profile takes the size of the support where it was started (the first click); supports of different sizes can be joined on the same line.

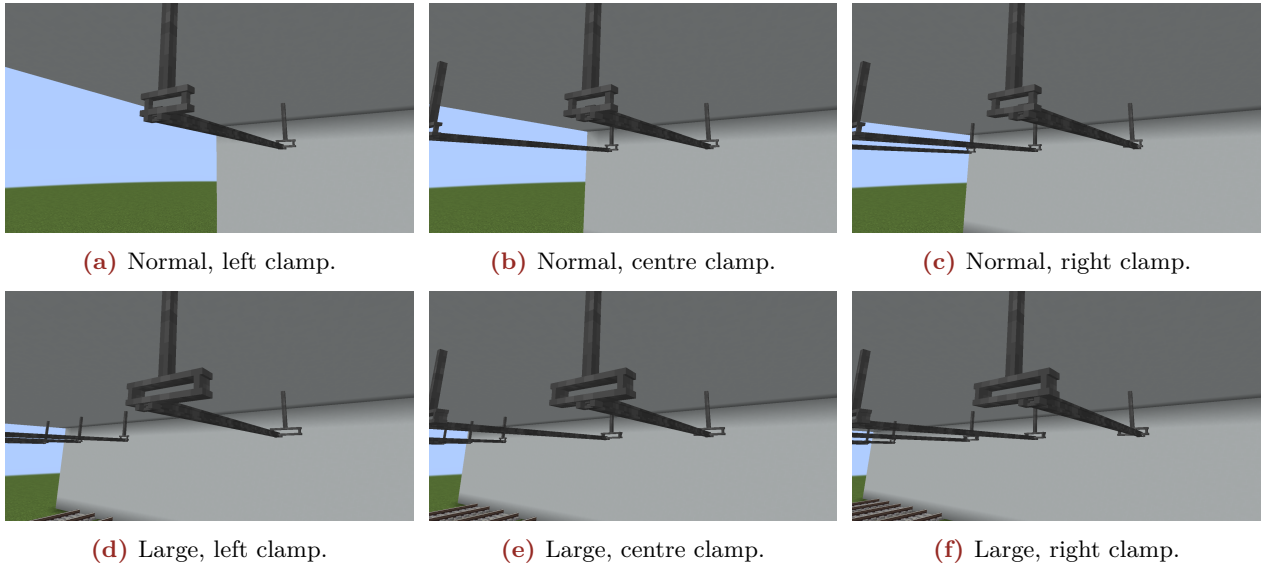


Figure 19: Ceiling supports: both sizes with the three clamp positions, each with its rigid profile.

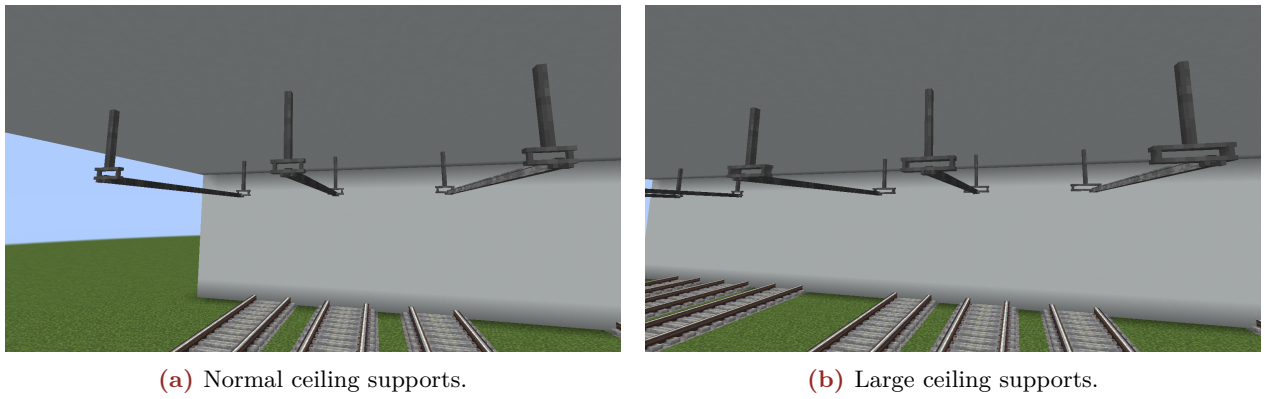


Figure 20: The three clamp positions side by side: left, centre and right.

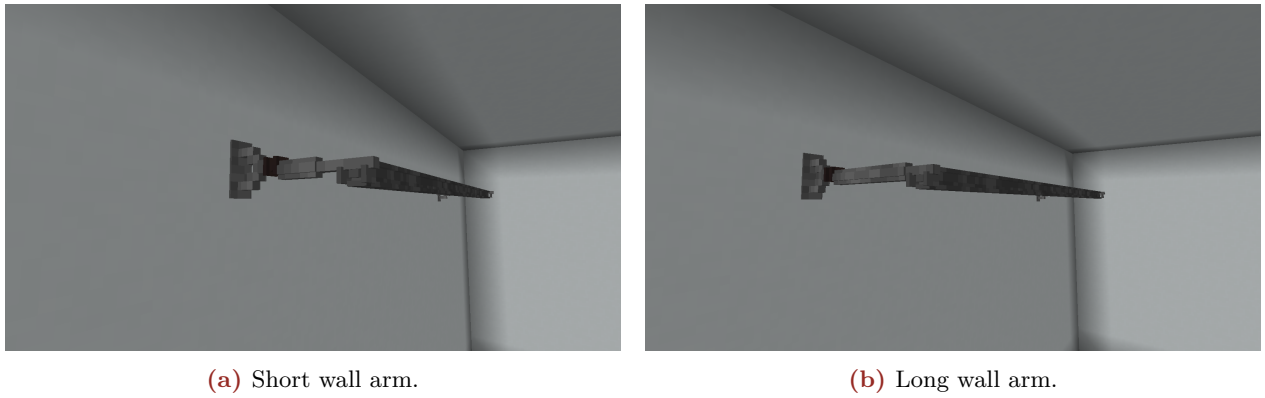


Figure 21: Wall version.

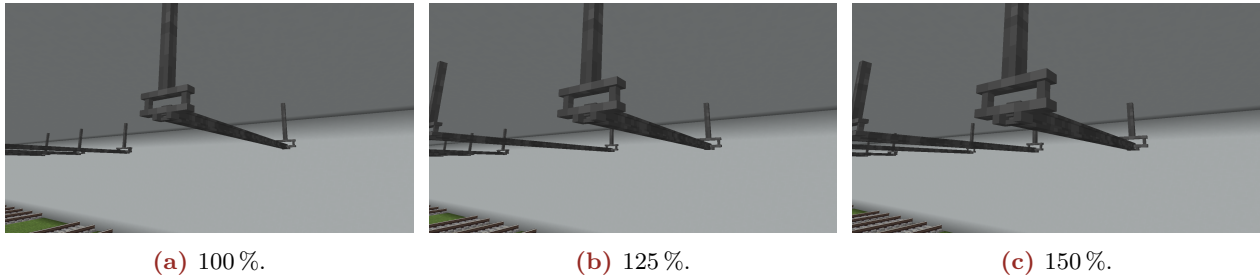


Figure 22: The normal ceiling support at the three overall sizes, with the profile at the same size.

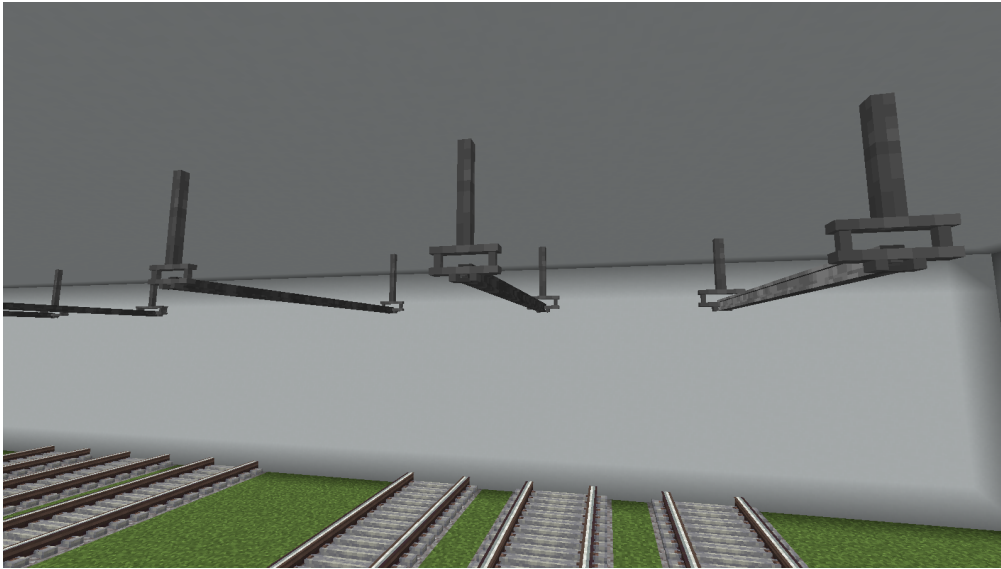
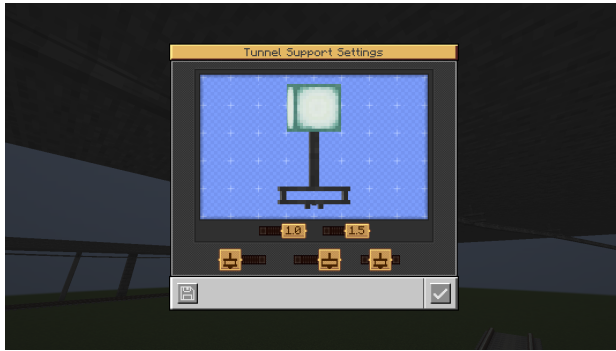


Figure 23: The three overall sizes side by side.

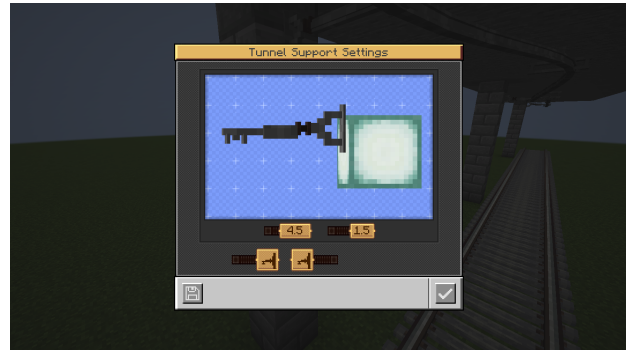
6.2 Settings window

Setting	Effect
Version	Ceiling or wall (also chosen automatically by the face that is clicked).
Size	Normal or large (ceiling); short or long arm (wall).
Clamp	Left, centre or right (ceiling only), to build the zigzag.
Height	Ceiling: how much longer the rod is. Wall: up or down from the centre of the block.
Overall size	100 %, 125 % or 150 % for the support and the profile strung from it.
Set default config	Stores the current settings as the default for new, unconfigured items.

As with the cantilever, the window opens with the item in hand (and the settings are kept in the item) or with an empty hand on a placed support; the wires already strung move with it.



(a) Ceiling version at 150 %, with the ceiling block shown for reference.



(b) Wall version at 150 %, with the wall block shown for reference.

Figure 24: Settings window of the tunnel support.

7 Iberian Pantograph

The Iberian Pantograph is PNW's pantograph with its frame painted red, as on many Spanish trains. It keeps the model, animation and behaviour of the original: it can be raised and lowered, it collects current from both the flexible and the rigid catenary, and it works on Create trains and contraptions. Insulators and the contact strip keep their original colours.

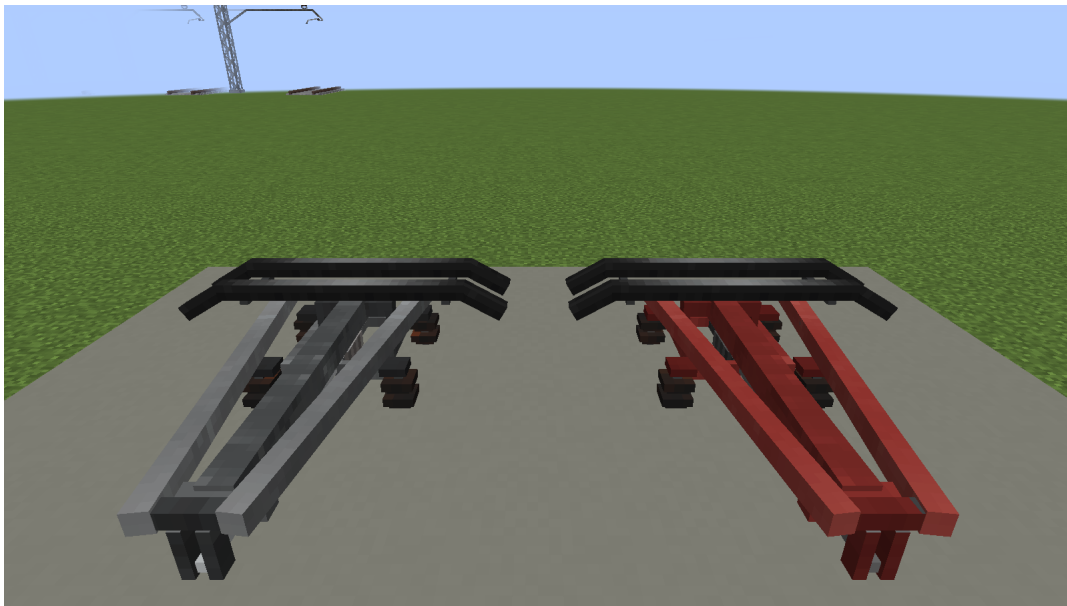


Figure 25: PNW pantograph (left) and Iberian Pantograph (right).

8 Items and crafting

	Item	Recipe
	Iberian Steel	Iron ingot and black dye (shapeless). Base material of the add-on.
	Iberian Cantilever	Create sequenced assembly from a PNW iron rod bundle: two loops of deploying Iberian Steel, deploying a brown insulator and pressing.
	Iberian Tunnel Support	Two Iberian Steel above a brown insulator; makes two.
	Rigid Catenary Profile Bundle	Six Iberian Steel.
	Iberian masts	PNW iron strips with Iberian Steel (and iron rods for the square mast); six per craft.
	Iberian Pantograph	PNW pantograph and red dye (shapeless).

All recipes are visible in recipe viewers such as JEI, including the Create sequenced assembly.

9 Technical summary

Minecraft	1.20.1, Forge 47
Required mods	Create: Pantographs & Wires (1.20.1-beta-0.2.3-C6), Create 6.0.8, DragonLib, GeckoLib
License	GPL-3.0, the same as Pantographs & Wires
Languages	English, Spanish, Catalan, German
Mod id	iberiancantilever

Integration with Pantographs & Wires The add-on relies only on PNW's public building blocks: wire connectors, wire types registered in PNW's wire registry and catenary graph, PNW's mast classes and tags, its pantograph and its GUI widgets. Its own content (models, textures, translations and code) is self-contained, so it can be distributed as a separate add-on that requires PNW, or merged into PNW as a set of additional blocks.

Credits *Create: Pantographs & Wires* by MrJulsen, on which this add-on is built, and *Create* by the Create team. Photographs from Wikimedia Commons, credited under each image and listed below.

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- *S01 882 Stromabnehmer BR 449* – Falk2, CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:S01_882_Stromabnehmer_BR_449.jpg

Changelog

- 8 October 2026**
 - **New:** diagonal stay for cantilevers of width 3 or more, strung with PNW's wire coil in *Support Wire* mode from the horizontal tube to a mast block 2 to 5 blocks higher. It is a single wire that enters the tube and the mast, its attachment moves outwards with the width, it follows the cantilever when it is reconfigured and it returns its wire when broken.
 - **New:** *Overall size* of the tunnel support (100 %, 125 %, 150 %), which enlarges the support and the rigid profile strung from it; sizes can be mixed on the same line.
 - **Fixed:** a tunnel support configured from the item lost its overall size when placed.
 - **Documentation:** new screenshots of every variant and this changelog.
- 4 October 2026**
 - German translation (the add-on is now in English, Spanish, Catalan and German).
 - Reference photographs of conventional Iberian catenary in this document.
- 3 October 2026**
 - Iberian Tunnel Support in ceiling and wall versions, two sizes each, with its settings window.
 - Rigid catenary bends through curves and joins the flexible catenary at tunnel mouths.
 - Iberian Pantograph.
 - New convex registration arm on middle and outer cantilevers; recipes visible in JEI.
 - First version of this document.
- 2 October 2026**
 - The add-on becomes *Iberian Cantilevers*: settings window in PNW's style with a 3D preview and per-player default settings.
 - Iberian masts (flat, flat diagonal and square lattice) with weathering and waxing.
 - Rigid tunnel catenary with its profile bundle; Iberian Steel and crafting recipes.
- 1 October 2026**
 - First version of the Iberian cantilever: three zigzag positions, three insulator types and optional support wire, with PNW's catenary wire attached.